# FPChecker Tutorial Example 6: Mathematical Formulation and Mixed-Precision Algorithmic Changes

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## 1 Purpose

This document describes the mathematics implemented by the three programs in Example 6:

- `fp32.cpp`: baseline FP32 algorithm,
- `mixed.cpp`: report-driven mixed-precision refinement,
- `fp64.cpp`: FP64 reference.

The goal is to show how high-relative-error lines in FP32 motivate both precision promotion and algorithmic changes.

## 2 Input and Generated Data

Given inputs

$$N, \quad A, \quad a,\ b,\ c,$$

the synthetic data sequence is

$$x_i = T + A\sin(0.01i) + 0.25(i \bmod 9), \quad i = 0, \ldots, N-1,$$

with trend constant

$$T = 10^8.$$

In the code, $A$ is `amp`. In `fp32.cpp` and `mixed.cpp`, the stored array uses `float`; in `fp64.cpp`, it uses `double`.

## 3 Quantity 1: Small Root of a Quadratic

The quadratic equation is

$$ar^2 + br + c = 0.$$

### 3.1 FP32 baseline (unstable form)

The FP32 program computes

$$\Delta = b^2 - 4ac,$$
$$r_{\text{small}}^{(\text{fp32})} = \frac{-b + \sqrt{\Delta}}{2a}.$$

When $b \gg c$ and $a \approx 1$, we get $\sqrt{\Delta} \approx b$, so

$$-b + \sqrt{\Delta}$$

subtracts nearly equal numbers. This is catastrophic cancellation, often driving the computed numerator to zero in FP32.

### 3.2 Mixed and FP64 (stable form via Vieta)

The mixed and FP64 programs compute the large root first:

$$r_{\text{large}} = \frac{-b - \sqrt{\Delta}}{2a},$$

and then use Vieta's product relation $r_{\text{small}} r_{\text{large}} = c/a$:

$$r_{\text{small}} = \frac{c}{a\, r_{\text{large}}}.$$

This avoids the unstable subtraction $(-b + \sqrt{\Delta})$ and is numerically much more robust.

## 4 Quantity 2: Variance of the Generated Data

### 4.1 FP32 baseline (one-pass formula)

Define

$$S_1 = \sum_{i=0}^{N-1} x_i, \qquad S_2 = \sum_{i=0}^{N-1} x_i^2, \qquad \mu = \frac{S_1}{N}.$$

The FP32 baseline returns

$$\sigma^2_{\text{one-pass}} = \frac{S_2}{N} - \mu^2.$$

For large-magnitude data (here dominated by $10^8$), both terms are large and close, so subtracting them causes severe cancellation.

### 4.2 Mixed and FP64 (two-pass formula)

First pass:

$$\mu = \frac{1}{N} \sum_{i=0}^{N-1} x_i.$$

Second pass:

$$\sigma^2_{\text{two-pass}} = \frac{1}{N} \sum_{i=0}^{N-1} (x_i - \mu)^2.$$

This form is better conditioned for the centered residuals and avoids subtracting two large nearly equal moments.

## 5 Algorithmic and Precision Changes from FP32 to Mixed

The mixed version is not only a type cast; it combines selective precision promotion with algorithm replacement.

| Component | FP32 baseline | Mixed version |
|---|---|---|
| Data storage $x_i$ | FP32 (float) | FP32 (float, unchanged) |
| Quadratic root | Unstable $(-b+\sqrt{\Delta})/(2a)$ in FP32 | Stable Vieta form in FP64 |
| Variance | One-pass $E[x^2]-E[x]^2$ in FP32 | Two-pass centered sum in FP64 |
| Accumulators | FP32 | FP64 |

## 6 Why These Specific Changes

FPChecker rounding-error reports in this example identify large relative errors at:

- the quadratic cancellation numerator $(-b+\sqrt{\Delta})$, and
- the one-pass variance subtraction $(S_2/N-\mu^2)$.

Hence mixed precision targets exactly those hotspots:

1. Promote sensitive arithmetic to FP64.
2. Replace unstable formulas with numerically stable equivalents.

Low-error operations (for example, data generation terms) remain FP32 to preserve the mixed-precision character.

## 7 Expected Accuracy Behavior

Relative to the FP64 reference:

- FP32 often has very large relative error for $r_{\text{small}}$ and for variance.
- Mixed should reduce these errors substantially.

In this example setup, the mixed result typically matches the FP64 quadratic small root closely and improves variance by many orders of magnitude compared with FP32.

## 8 Build Instructions

From the `tutorial/example_6` directory:

```
pdflatex example6_math_formulas.tex
```

If you prefer DVI first:

```
latex example6_math_formulas.tex
```